

required coverage for DO-254 based DAL A, B and C

Question 01:

what is the required coverage for DO-254 based DAL A, B and C

In **DO-254** (Design Assurance Guidance for Airborne Electronic Hardware), the required **coverage levels** depend on the **Design Assurance Level (DAL)** assigned to the hardware function. Here's a breakdown of the **typical verification coverage expectations** for DAL A, B, and C:

✓ 1. DAL A (Most Critical – Catastrophic failure conditions)

- **Coverage Requirements:**
 - **100% Requirements Coverage** – All high-level and low-level requirements must be verified.
 - **100% Code Coverage:**
 - **Statement Coverage**
 - **Branch Coverage**
 - **Modified Condition/Decision Coverage (MC/DC)** where applicable (especially for complex logic).
 - **Full Traceability** from requirements to design, to implementation, to verification.
 - **Robustness Testing** (e.g., stress, corner-case, and error conditions).
 - **Formal Verification** or other rigorous methods often recommended for complex FPGA/ASIC logic.

✓ 2. DAL B (Hazardous/Severe-Major failure conditions)

- **Coverage Requirements:**
 - **100% Requirements Coverage**

- **High Code Coverage:**
 - **Statement Coverage**
 - **Branch Coverage**
 - **MC/DC** is **not required**, but good practice in critical paths.
 - **Traceability** from requirements to implementation and tests.
 - **Robustness Testing** should be done but can be less exhaustive than DAL A.
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✓ 3. DAL C (Major failure conditions)

- **Coverage Requirements:**
 - **Requirements Coverage** is still required but:
 - Less rigorous than DAL A/B.
 - Focus is on **functional verification** and **basic structural coverage**.
 - **Code Coverage:**
 - **Statement Coverage** is typically sufficient.
 - **Branch or MC/DC** is **not required**.
 - **Traceability** is expected but can be limited to **high-level requirements**.
 - **Less formal verification** is acceptable.
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Source: <https://chatgpt.com/c/6889158a-1630-8011-876f-96128717851e>

Summary Table

Category	DAL A	DAL B	DAL C
Requirements Coverage	100%	100%	100% (less depth)
Code Coverage	Statement, Branch, MC/DC	Statement, Branch	Statement only
Traceability	Full (all levels)	Full (all levels)	Partial (HLS okay)
Robustness Testing	Required (extensive)	Required (moderate)	Optional
Formal Methods	Encouraged	Optional	Not required

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Let me know if you'd like a checklist or template for a DO-254 verification matrix.

You said:

yes

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Here is a **DO-254 Verification Matrix Template** (DAL A/B/C compliant) — formatted as a table for tracking **requirements, implementation, and verification coverage**:

✓ DO-254 Verification Matrix Template

ID	Requirement Description	DAL	Design Artifact	Implementation Ref	Verification Method	Test Case ID	Coverage Type	Passed? (Y/N)	Comments
RQ-001	The system shall reset on power-up	A	RTL block diagram	reset_controller.vhd	Simulation, Review	TC-001	Statement, Branch, MC/DC	Y	Full reset verified
RQ-002	SPI interface shall support 8-bit transfers	B	SPI module design	spi_ctrl.vhd	Simulation, RTL Code Review	TC-010	Statement, Branch	Y	Timing met
RQ-003	ADC input must be sampled every 1 ms	C	ADC timing spec	adc_timer.vhd	Simulation	TC-020	Statement	Y	No anomalies
...	...	...	...	...	...	...	...	...	...

Column Descriptions

Column	Purpose
ID	Unique identifier for each requirement (e.g., RQ-001)
Requirement Description	Text of the hardware or functional requirement
DAL	Assurance Level (A, B, or C)
Design Artifact	Design element where this requirement is implemented (block name, schematic, etc.)
Implementation Ref	Specific VHDL/Verilog module or document
Verification Method	Simulation, Testbench, Review, Analysis, etc.
Test Case ID	Unique ID for test case validating this requirement
Coverage Type	What kind of code coverage is needed (e.g., Statement, Branch, MC/DC)
Passed?	Test result (Yes/No)
Comments	Notes on any anomalies, waivers, or issues found